

Artificial Intelligence

Expanded Knowledge Base Document for RAG / Document Question-Answering

1. Introduction to Artificial Intelligence

Artificial intelligence (AI) is the field of computing concerned with building systems that can perform tasks that normally require aspects of human intelligence. These tasks include perception, reasoning, learning, language processing, planning, and decision support. AI systems may use explicit rules, statistical methods, machine learning, neural networks, or combinations of these techniques.

2. Major Areas of AI

AI includes several connected areas. Perception deals with interpreting images, audio, sensor readings, and other signals. Reasoning focuses on drawing conclusions from information. Planning selects actions needed to achieve a goal. Learning allows systems to improve from examples. Natural language processing enables computers to work with human language, while computer vision focuses on visual information. Robotics combines perception, reasoning, planning, and physical action.

3. Rule-Based and Learning-Based Systems

Traditional rule-based systems encode explicit instructions. They can be useful when the rules of a domain are stable and clearly understood. Machine learning systems instead learn patterns from examples. A learning system generally requires data, an objective, a model or algorithm, and an evaluation procedure. The choice between approaches depends on the problem, available data, constraints, and required reliability.

4. AI Applications

AI applications include recommendation systems, fraud detection, search engines, computer vision, speech recognition, robotics, medical decision support, and document analysis. In a recommendation system, a model can use historical interactions to estimate which items may be relevant. In document analysis, AI can extract information, classify documents, summarize content, or answer questions from a collection of files.

5. Data, Models, and Evaluation

An AI system typically involves data, an algorithm or model, an evaluation procedure, and an environment in which predictions or decisions are used. Data quality is important because incomplete, inconsistent, or poorly represented data can reduce system performance. Evaluation should use measures appropriate to the task and should distinguish training data from data used to assess generalization.

6. Responsible AI Engineering

Good AI engineering requires attention to data quality, security, privacy, monitoring, and responsible deployment. A model that performs well in a controlled experiment may behave differently after deployment because the environment or input data changes. Monitoring can help identify performance changes, unusual inputs, and operational problems. Privacy considerations are particularly important

when systems process personal or sensitive information.

7. AI Project Lifecycle

A practical AI project can be organized into problem definition, data collection, preparation, model development, evaluation, deployment, and monitoring. Problem definition clarifies what the system should predict or produce. Data preparation makes the information suitable for the selected method. Evaluation checks whether the system meets its requirements. Deployment makes the model available to users, while monitoring provides feedback for maintenance and improvement.

8. AI and Other Fields

Artificial intelligence is an umbrella area that includes machine learning and overlaps with deep learning, natural language processing, computer vision, robotics, and information retrieval. Machine learning focuses on algorithms that learn patterns from data. Deep learning uses neural networks with multiple layers. NLP addresses computational work with human language. These areas can be combined in systems such as document question-answering applications.

9. Key Takeaways

AI systems are not defined by one algorithm. They are engineered systems that connect data, models, evaluation, and an operating environment. Effective AI development therefore requires both technical methods and attention to reliability, security, privacy, and deployment conditions.